

Checklist A: Decision Latency Capture Sheet

1. Purpose and Scope

This instrument measures Decision Latency (DL), defined as the elapsed time between the moment actionable incident intelligence becomes observable to a responsible operator and the moment a first valid technical response action is executed. Decision Latency isolates human cognitive processing time from technical detection latency, providing a diagnostic explanation for containment timing outcomes observed in IRRS scoring.

This sheet is completed by the designated evaluator during or immediately after a simulation. It is not a performance assessment of individuals. All recorded values reflect system-level organisational behaviour under simulated conditions and are used exclusively for diagnostic and improvement purposes.

This instrument informs the following IRRS sub-metrics: Containment-action timing, Decision flow, and Role clarity.

2. Pre-Simulation Setup

Complete the following fields before the simulation begins.

Organisation / Team	
Simulation Date	
Scenario Class	<i>Ransomware / Insider Threat / Cloud Misconfiguration / Credential Leakage (circle one)</i>
SRI Level	<i>1 / 2 / 3 / 4 / 5 (circle one)</i>
Simulation Mode	<i>Tabletop / Live Emulation (circle one)</i>
Evaluator Name	

3. Valid Action Rules

Only injects that require a non-trivial decision, such as containment, escalation, or credential revocation, are instrumented. Not all alerts qualify. A valid action must satisfy both of the following criteria:

- The action must materially affect the trajectory of the incident response, for example containment, investigation, or recovery actions.
- The action must be technically executable, not a discussion, a planning statement, or a request for approval.

Valid Actions (count these) Isolate endpoint from network Disable compromised user account Apply USB restriction policy Revoke cloud access token Initiate formal escalation to incident commander	Not Valid Actions (do not count these) "Let us double-check before we act" "Should we escalate this?" Verbal discussion of options with no action taken Reviewing logs without issuing a response command Waiting for additional confirmation before acting
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4. Decision Latency Recording Table

Record one row per decision-critical inject. T visible is the timestamp at which actionable intelligence is presented to the operator. T action is the timestamp of the first valid technical response action. DL is the difference in minutes. Record all delays without judgement; the Observed Friction column captures the qualitative reason for the delay and is used diagnostically, not punitively.

Inject ID	Event Description	T visible (hh:mm)	T action (hh:mm)	DL (mins)	Observed Friction / Notes
Mean DL (sum of DL values / number of injects):					Max DL observed:

5. Scenario Threshold Guidance

The following reference bands indicate when a calculated Decision Latency value should be considered acceptable, elevated, or critical. These thresholds are practitioner-informed reference points calibrated to scenario class and SRI level. They are intended to support diagnostic interpretation and should be refined through repeated simulation cycles as empirical data accumulates.

Important: The same latency value carries different diagnostic weight depending on scenario class. A 10-minute Decision Latency in an active ransomware scenario (SRI 5) is classified as critical because rapid containment is paramount to preventing lateral spread. The same duration in an insider threat investigation (SRI 4) may fall only within the elevated range, as deliberate verification before action is operationally appropriate and premature containment may compromise the investigation. Always interpret DL values relative to the scenario class and SRI level recorded in Section 2.

Scenario Class	SRI	Acceptable	Elevated	Critical
Ransomware Propagation	SRI 5	Under 5 mins	5 to 10 mins	Over 10 mins
Insider Data Exfiltration	SRI 4	Under 10 mins	10 to 15 mins	Over 15 mins
Cloud Misconfiguration	SRI 3	Under 15 mins	15 to 25 mins	Over 25 mins
Credential Leakage	SRI 3	Under 15 mins	15 to 25 mins	Over 25 mins

After calculating the mean DL and maximum DL, use the table above to assign a classification for each inject. Where the mean and maximum fall into different bands, record both and note the discrepancy in the evaluator summary.

6. Worked Example (Insider Threat, SRI 4)

The following completed example is drawn from the theoretical worked example presented in the IRRS paper. It illustrates how the recording table is used in practice and how the threshold guidance is applied to assign a classification.

Inject	Event Description	T visible	T action	DL (mins)	Observed Friction
1	Unusual file access spike detected	09:02	09:09	7	Analyst sought verbal approval before acting
2	Evidence of removable media usage confirmed	09:18	09:30	12	Uncertainty about whether USB restriction would disrupt operations
3	Data confirmed copied to personal cloud storage	09:41	09:55	14	Senior manager consulted before token revocation; no clear pre-authorisation existed
Mean DL = $(7 + 12 + 14) / 3 = 11.0$ mins Max DL = 14 mins				Classification: ELEVATED (SRI 4, Insider Threat)	

In this example, the mean DL of 11.0 minutes falls within the Elevated band for an SRI 4 scenario. The maximum DL of 14 minutes approaches the Critical threshold. The primary friction cause across all three injects was the absence of pre-authorised decision triggers, forcing the team to seek approval before each containment action. This directly explains the low score in the Containment-action timing sub-metric ($si = 3$, $wi = 4$) in the corresponding IRRS calculation.

7. Evaluator Guidance Notes

The following principles should guide evaluators when completing this instrument:

- Record T visible and T action as accurately as possible. In tabletop simulations, use the facilitator inject log as the authoritative source for T visible. In live emulations, use alert timestamps from the SIEM or adversary emulation platform.
- Where a team takes multiple actions in response to a single inject, record only the first valid action as T action. Subsequent actions may be noted in the Observed Friction column.
- If a team fails to take any valid action in response to an inject, record the end of the simulation phase as T action and note the absence of response explicitly.
- Friction observations should describe the systemic or procedural cause of a delay where visible, for example, unclear escalation authority, absence of pre-authorised procedures, or verification loops. Do not attribute delays to named individuals.
- This instrument is designed for system-level diagnostic analysis only and must not be used to evaluate individual competence or performance.

This instrument is part of the IRRS Human-Centric Telemetry suite. It should be used alongside Checklist B (Communication Entropy Tally Sheet) and Checklist C (Authority Drift Tracker) to provide a complete diagnostic picture of human-system interaction during the simulation.